A NON-MF GROUP

SAUERS

Abstract. A countable discrete group is MF if it embeds in the unitary group of a norm matrix corona $\mathcal{Q} = \prod_n M_{d_n}(\mathbb{C})/\bigoplus_n M_{d_n}(\mathbb{C})$. We prove that non-MF groups exist, resolving in the negative the longstanding question of whether every countable group is MF. Our principal example is an explicit finitely presented group.

At the final obstruction step, we adapt the operator–Hilbert–Schmidt compression argument of Bachner–Dogon–Lubotzky for Deligne-type groups. A key ingredient is a one-sided Kazhdan transport theorem: in every operator-norm asymptotic representation, one-sided conjugation preserves the Hilbert–Schmidt asymptotic commutant of a property-(T) subgroup. This forces every homomorphism into the unitary group of a norm matrix corona to map a distinguished central involution to the identity, while an auxiliary Clifford quotient shows that the involution is nontrivial.

2020 *Mathematics Subject Classification.* Primary 46L05; Secondary 20F05, 22D25, 20E26.

Key words and phrases. MF groups, approximate unitary representations, property (T), group C^*-algebras, sofic groups.

Contents

1. Introduction

The *MF problem* (MF = matricial field) asks whether every countable discrete group admits finite-dimensional matrix models whose multiplicative errors tend to zero in operator norm and which asymptotically separate every nontrivial element from the identity [Sh]. The separation clause is part of the definition, as constant identity maps are asymptotically multiplicative for every group. Throughout, MF for groups means the operator-norm property of Carrión–Dadarlat–Eckhardt [CDE]; stronger conventions are discussed after Proposition 2.3. To our knowledge no counterexample was previously known [Sc, Sh]. We construct an explicit eight-generator, forty-one-relator group that is not MF. The presentation uses separate generators for the three translations and the three generators of the linear action. The Tietze reduction in Section 7 gives a six-generator, thirty-two-relator presentation.

Residually finite and LEF groups are MF [CDE]. Amenable groups are MF because their reduced group C^*-algebras embed into the universal UHF algebra [Sc20, Theorem B]. The underlying quasidiagonality input is the Tikuisis–White–Winter theorem, in particular their proof of Rosenberg's conjecture [TWW].

Several nearby notions are stronger. Schafhauser's group models additionally approximate the left regular representation [Sc], and the strong models of Gao–Kunnawalkam Elayavalli–Manzoor–Patchell recover the regular norm [GKEMP]. Shulman's amalgamated-product results concern, in particular, the MF property of full group C^*-algebras [Sh]. Each of these stronger group or C^*-algebraic properties implies the group property used here.

The result.

Theorem A (an explicit finitely presented non-MF group)**.** *Let E be the group on the eight generators $v_1, v_2, v_3, x, y, z, t, c$ with the forty-one relators displayed in Definition 6.1, and put*

$$\hat{c} = tct^{-1}, \qquad u = [\,\hat{c}, v_1], \qquad w = u^2 = [\,\hat{c},\, v_1\hat{c}v_1^{-1}].$$

The element w is central and satisfies $w^2 = 1$.

1. *$w \neq 1$ in E, so w is a central involution;*
2. *for every sequence (d_n) of positive integers and every homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q}((d_n)))$ into the unitary group of the corresponding matrix quotient, one has $\Theta(w) = 1$.*

In particular E is not MF, and neither $C^*_{\max}(E)$ nor $C^*_{\mathrm{r}}(E)$ is an MF C^*-algebra.

Further variants and consequences are collected separately in *Non-MF group notes*.

Outline of the proof. The final analytic maneuver is adapted from the operator–Hilbert–Schmidt compression argument of Bachner–Dogon–Lubotzky [BDL, Proposition 1.5]. Starting from a separated central involution, their proof passes to its negative spectral corner and uses operator–Hilbert–Schmidt stability to obtain nearby genuine finite-dimensional representations. Here the same corner reduction is combined with a different source of Hilbert–Schmidt rigidity: one-sided conjugation of a property-(T) subgroup.

Multiplicative defects are measured in operator norm; asymptotic-commutant estimates are measured in normalized Hilbert–Schmidt norm. Let $L \le H$ have property (T), and let $t, c \in H$ satisfy

$$tLt^{-1} \subseteq L, \qquad [c, L] = 1.$$

Conjugation by the unitaries in an operator-norm asymptotic representation (U_n) of H defines an asymptotic representation on the Hilbert–Schmidt spaces $L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$. This is where the operator norm enters: it controls the multiplicative defects of the conjugation actions. Property (T) produces the projection onto the L-invariant vectors. The subgroup inclusion implies that this projection is dominated by its conjugate under t. The two projections are equivalent in a finite algebra and therefore equal. Consequently, conjugation by t preserves the asymptotic commutant of L in normalized Hilbert–Schmidt norm (Theorem 3.5).

Put $\hat{c} = tct^{-1}$ and $u = [\,\hat{c}, a\,]$ for some $a \in L$. Since c commutes with L, the sequence $(U_n(c))$ lies in that asymptotic commutant. Theorem 3.5 therefore gives $\|[U_n(\hat{c}), U_n(\ell)]_-\|_2 \to 0$ for every $\ell \in L$. Consequently, in every operator-norm asymptotic representation (U_n) of H,

$$\|U_n(u) - 1\|_2 \longrightarrow 0.$$

Suppose now that $w = u^2$ is central in H with $w^2 = 1$, and let Θ be a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $q = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection commuting with $\Theta(H)$. Compression to the corresponding matrix corners, equipped with their normalized traces, gives an operator-norm asymptotic representation of H in which the image of w converges to -1 in operator norm. There the display above gives $\|U_n(u)^2 - 1\|_2 \to 0$, while $\|-1-1\|_2 = 2$. Hence $\Theta(w) = 1$, and if $w \neq 1$ then H is not MF (Theorem 4.2). Normalizing the trace on each corner is essential: enlarging a model by an identity block preserves operator-norm defects and separations while making normalized Hilbert–Schmidt distances arbitrarily small.

Let Γ be a finitely presented Kazhdan group with a proper injective endomorphism α, let $a \in \Gamma \setminus \alpha(\Gamma)$, and adjoin t and c subject to $t\gamma t^{-1} = \alpha(\gamma)$, $c^2 = 1$ and $[c, \gamma] = 1$. In the ascending HNN extension of Γ along α the cosets $t\Gamma$ and $at\Gamma$ are distinct, since $t^{-1}at \in \Gamma$ would give $a \in \alpha(\Gamma)$. An auxiliary Clifford quotient witnesses that $w \neq 1$. For a set X, the Clifford group $\mathrm{Cl}(X)$ is generated by a central involution ζ together with involutions c_x, $x \in X$, subject to $[c_x, c_y] = \zeta$ for $x \neq y$; every permutation of X acts on it by permuting the c_x and fixing ζ (Construction 8.1). Let the ambient group act in this way on $\mathrm{Cl}(X)$ for X the coset space, and send c to the generator c_x at the base coset. Then $\hat{c}$ and $a\hat{c}a^{-1}$ go to the generators at the two distinct cosets, so u goes to a product of two anticommuting involutions and $w = u^2$ to $\zeta \neq 1$ (Theorem 5.1).

Taking

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \qquad \alpha(v, A) = (2v, A), \qquad a = (e_1, I)$$

gives the group E of Theorem A.

Property (T) is used only for the embedded base subgroup Γ: the group E itself maps onto $\mathbb{Z}$ by sending t to 1 and every other generator to 0, and therefore does not have property (T). An exact finite-dimensional counterpart of the commutant argument, with equality of dimensions in place of finiteness of the ultraproduct, is Theorem 3.3.

Relation to prior work. *MF algebras and traces.* The MF problem for C^*-algebras predates the group question. Blackadar and Kirchberg, who introduced MF algebras, remarked that no stably finite separable C^*-algebra was known not to be MF [BK, CDE]. Their natural candidate was $C^*_{\mathrm{r}}(F_2)$, which Haagerup and Thorbjørnsen later proved to be MF [HT]. The negative solution of the Connes embedding problem subsequently implied the existence of abstract stably finite non-MF algebras [JNVWY, FGH]. Theorem A gives an explicit reduced group C^*-algebra that is not MF.

Metric approximation problems. In his 2018 ICM address, Thom asked for which Schatten p-norms there exist non-approximable groups [Th, Question 3.11]. The case $p = 2$ was settled by De Chiffre, Glebsky, Lubotzky, and Thom [DGLT], the cases $1 < p < \infty$ by Lubotzky and Oppenheim [LuO], and the case $p = 1$ by Bachner, Dogon, and Lubotzky [BDL]. Those results left three approximation problems open: whether every countable group is sofic (permutations in the Hamming metric), hyperlinear (unitaries in the normalized Hilbert–Schmidt norm), or MF (unitaries in operator norm). The first was answered negatively in 2026: a nonsofic group was constructed in [OAI], Fournier-Facio then gave a torsion-free example [FFF], and Kun–Thom developed further wreath-product examples [KT26]. Theorem A settles the MF problem negatively; as of August 2026 the hyperlinear problem remains open. The nonsofic and MF arguments both use a property-(T) subgroup Γ satisfying $t\Gamma t^{-1} \subseteq \Gamma$. In the MF setting, conjugation by t preserves the Hilbert–Schmidt asymptotic commutant of Γ whenever the multiplicative defects converge to zero in operator norm.

Related arguments. Recent work on nonsofic groups uses one-sided conjugation of a property-(T) subgroup together with an element centralizing that subgroup [OAI, KT26], building on rigidity results of Kun and Kun–Thom [Ku16, KT19]. Alekseev–Thom prove a related centralizer rigidity for sofic embeddings [AT]. Those results concern Hamming or tracial approximations. In the present operator-norm setting, the group acts by conjugation on the Hilbert–Schmidt space of each matrix block. Finiteness of the norm ultraproduct implies that the projection onto the property-(T) subgroup's fixed subspace equals its unitary conjugate (Theorem 3.5).

Slofstra's hyperlinear-profile construction uses a Clifford group, a distinguished central involution, a shift, and an HNN doubling map [Slo18, Section 2 and Remark 3.3]. He also considers the elements sent to the identity by every exact finite-dimensional representation and those sent to the identity by every approximate representation [Slo17, Definitions 2.5–2.7]. These subgroups are the exact and approximate analogues of the MF kernel considered here.

Bekka and Bekka–Valette study property (T) through the conjugation representation $\pi \otimes \bar{\pi}$ on Hilbert–Schmidt space [Be, BV]. Dadarlat uses normalized finite-rank projections as almost invariant Hilbert–Schmidt vectors in an operator-norm approximation argument [Dad, Lemma 3.18 and Proposition 3.19]. The group-MF framework is that of Carrión–Dadarlat–Eckhardt, who also give, through Abels' group, an amenable MF group that is not residually finite [CDE, Section 2.14 and Corollary 2.18].

If $e\colon A \to \mathcal{Q}$ is an MF embedding of a unital group C^*-algebra and $p = e(1)$, then $g \mapsto e(u_g) + (1-p)$ is an MF embedding of the underlying group. The converse is open: an injective homomorphism $G \to \mathcal{U}(\mathcal{Q})$ is not known in general to induce a C^*-algebra embedding $C^*_{\mathrm{r}}(G) \hookrightarrow \mathcal{Q}$. Theorem A rules out both embeddings for E.

2. Matrix quotients

All groups are discrete and countable. We use the standard formulation of property (T) for unitary representations on complex Hilbert spaces. For group elements $[g,h] = ghg^{-1}h^{-1}$; for operators $[x,y]_- = xy - yx$. Throughout, $*$-homomorphisms are not assumed unital unless explicitly stated; in particular, an MF embedding may be nonunital. For a unital C^*-algebra A we write $\mathcal{U}(A)$ for its unitary group. On $M_r(\mathbb{C})$ we use the normalized trace tr_r, the operator norm $\|\cdot\|$, and the normalized Hilbert–Schmidt norm $\|x\|_2 = \mathrm{tr}_r(x^*x)^{1/2}$. For a nonzero projection $q \in M_r(\mathbb{C})$ we equip the corner $qM_r(\mathbb{C})q$ with its own normalized trace. When the trace must be specified, we write $\|x\|_{2,\mathrm{tr}_r}$. The unnormalized trace is $\mathrm{Tr} = r \cdot \mathrm{tr}_r$ on $M_r(\mathbb{C})$. We repeatedly use the inequalities $\|x\|_2 \le \|x\|$; $|\mathrm{tr}_r(x)| \le \|x\|$; and $\|uxv\|_2 = \|x\|_2$ for unitaries u, v.

Given a sequence $(d_n)_{n\in\mathbb{N}}$ of positive integers, write

$$\mathcal{Q}((d_n)_{n\in\mathbb{N}}) \;=\; \prod_{n\in\mathbb{N}} M_{d_n}(\mathbb{C}) \Big/ \bigoplus_{n\in\mathbb{N}} M_{d_n}(\mathbb{C}), \tag{1}$$

where the product is the bounded C^*-product and $\bigoplus_n$ is the c_0-sum, the ideal of sequences with $\|x_n\| \to 0$; equivalently, $\mathcal{Q}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$. Write $q\colon \prod_n M_{d_n}(\mathbb{C}) \to \mathcal{Q}$ for the quotient map.

Following Blackadar–Kirchberg [BK], a separable C^*-algebra is *MF* if it embeds into an algebra of the form (1).

Convention on the dimension sequence. When the dimension sequence is not displayed, $\mathcal{Q}$ denotes $\mathcal{Q}((d_n)_{n\in\mathbb{N}})$ for a sequence fixed by the surrounding discussion. If no sequence has been fixed, a map into $\mathcal{Q}$ or $\mathcal{U}(\mathcal{Q})$ is understood to allow the sequence (d_n) to vary; likewise, a quantification over all such maps also ranges over all dimension sequences. Proposition 2.3 shows that nothing changes if the dimensions are required to be strictly increasing.

Three quotients of $\prod_n M_{d_n}(\mathbb{C})$ appear below, differing in which sequences they annihilate:

- $\mathcal{Q}$, where $\|x_n\| \to 0$: the norm matrix corona (1), and the target of an MF embedding;
- B_ω, where $\lim_\omega \|x_n\| = 0$: the norm ultraproduct of Section 3, a finite C^*-algebra, which contains the Kazhdan projection;
- the tracial ultraproduct, where $\lim_\omega \|x_n\|_2 = 0$: the target for hyperlinearity.

The ideals defining $\mathcal{Q}$ and B_ω are given by operator-norm convergence, the third by normalized Hilbert–Schmidt convergence. The operator norm is what makes the adjoint actions below asymptotically multiplicative on Hilbert–Schmidt space.

The next two lemmas are standard (cf. [BK, Lo98]); we include the short proofs to fix the constants used later.

Lemma 2.1 (lifting)**.** *Every unitary $u \in \mathcal{Q}$ lifts to a sequence of unitaries.*

Proof. Lift u to a bounded sequence (x_n). Unitarity of $q((x_n))$ gives $\|x_n^* x_n - 1\| \to 0$. Once $\|x_n^* x_n - 1\| \le \frac{1}{2}$, the matrix x_n is invertible and the polar correction $u_n = x_n (x_n^* x_n)^{-1/2}$ is unitary with $\|u_n - x_n\| \le 2\,\|x_n\|\,\|x_n^* x_n - 1\| \to 0$ by continuous functional calculus. Set $u_n = 1$ at the finitely many remaining indices. $\square$

Write

$$\mathcal{N} = \{(u_n) \in \textstyle\prod_n \mathcal{U}(d_n) \ : \ \|u_n - 1\| \longrightarrow 0\}, \tag{2}$$

a normal subgroup of $\prod_n \mathcal{U}(d_n)$.

Lemma 2.2 (unitaries in the quotient)**.** *For every sequence (d_n), with $\mathcal{N}$ as in (2), the map*

$$\kappa_{(d_n)} \colon \prod_n \mathcal{U}(d_n)/\mathcal{N} \longrightarrow \mathcal{U}(\mathcal{Q}), \qquad [(u_n)] \longmapsto q((u_n)),$$

is a canonical group isomorphism. Thus a homomorphism $H \to \mathcal{U}(\mathcal{Q})$ may be represented by unitary lifts $U_n(g)$ whose multiplicative defects tend to 0 in operator norm, with two lift sequences identified when their quotient tends to 1.

Proof. The kernel of $(u_n) \mapsto q((u_n))$ is exactly $\mathcal{N}$, so the induced map is injective. Surjectivity is Lemma 2.1. $\square$

An *operator-norm asymptotic representation* of H is a sequence of maps $U_n \colon H \to \mathcal{U}(d_n)$ such that $\|U_n(g)U_n(h) - U_n(gh)\| \to 0$ for all $g, h \in H$. Taking $g = h = 1$ gives $\|U_n(1) - 1\| \to 0$ automatically, since $U_n(1)$ is unitary. By Lemma 2.2, choosing unitary lifts $U_n(g)$ of $\Theta(g)$ turns a homomorphism

$\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ into such a sequence, and every such sequence arises this way. A countable group is *MF* if it admits an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ for some sequence of dimensions.

For a finite set $F \subseteq H$, a separation constant $\delta > 0$, and a tolerance $\varepsilon > 0$, an *approximate unitary representation* is a map $V\colon H \to \mathcal{U}(r)$, for some $r \geq 1$, such that

$$\begin{aligned} \|V(g)V(h) - V(gh)\| &\leq \varepsilon \quad (g, h \in F), \\ \|V(g) - V(h)\| &\geq \delta \quad (g \neq h \in F). \end{aligned}$$

Proposition 2.3 (Equivalent formulations of the MF property)**.** *For every countable group, the group-MF definition above — an injective homomorphism to $\mathcal{U}(\mathcal{Q})$ — is equivalent to the definition by sequences of unitaries. Both are equivalent to the existence of approximate unitary representations with separation constant* 1*, and allowing arbitrary positive dimensions is equivalent to requiring the dimensions to be strictly increasing.*

The nontrivial point in this equivalence is that an element-dependent positive separation in the corona can be made uniform, by the following operator-norm amplification.

Lemma 2.4 (tensor amplification)**.** *Let $u, v \in \mathcal{U}(d)$ and $\|u - v\| \geq \delta > 0$. If $N \in \mathbb{N}$ satisfies $8 < N\delta^2$, then for some $1 \leq p \leq N$,*

$$\left\|u^{\otimes p} - v^{\otimes p}\right\| \geq 1.$$

Proof. Put $z = u^*v$. It has a spectral value $e^{2\pi i\alpha}$ with $|1 - e^{2\pi i\alpha}| \geq \delta$. If $\alpha' = \min(\{\alpha\}, 1 - \{\alpha\})$, then $2\sin(\pi\alpha') \geq \delta$. Successive multiples move by $\pm\alpha'$ modulo 1, so some $p \leq \lceil 1/(6\alpha')\rceil \leq \pi/(3\delta) + 1$ has $\{p\alpha\} \in [1/6, 5/6]$. Since $e^{2\pi ip\alpha}$ belongs to the spectrum of $z^{\otimes p}$,

$$\left\|u^{\otimes p} - v^{\otimes p}\right\| \geq |1 - e^{2\pi ip\alpha}| \geq 1.$$

Finally, $\delta \leq 2$ for two unitaries, and $8 < N\delta^2$ therefore gives $\pi/(3\delta) + 1 < N$. ☐

Telescoping the difference of two p-fold tensor products of contractions shows that a pth tensor power multiplies every multiplicative defect by at most p.

Proof of Proposition 2.3. Lemma 2.2 gives the equivalence between a homomorphism to $\mathcal{U}(\mathcal{Q})$ and an operator-norm asymptotic representation. Positive dimensions can be made strictly increasing by replacing the nth model V_n with

$$I_{d_0} \oplus \cdots \oplus I_{d_{n-1}} \oplus V_n.$$

Block-diagonal operator norms are maxima, so this changes neither defects nor separations.

Suppose $\rho\colon G \to \mathcal{U}(\mathcal{Q}((d_n)))$ is injective and choose unitary lifts $u_n(g)$. Fix a finite $F \subseteq G$ and $\varepsilon > 0$. For each $g \neq h$ in F, injectivity supplies

$\delta_{g,h} > 0$ and arbitrarily large coordinates with $\|u_n(g) - u_n(h)\| \geq \delta_{g,h}$. Choose $N_{g,h}$ with $8 < N_{g,h}\delta_{g,h}^2$, and choose such a coordinate where every required multiplicative defect is at most $\varepsilon/N_{g,h}$. Lemma 2.4 gives a tensor power $p \leq N_{g,h}$ which separates this pair by at least 1, while every required defect remains at most ε. The block sum of these pair-specific models has defect at most ε and separates every distinct pair in F by at least 1. If $|F| \leq 1$, the constant one-dimensional model suffices.

Conversely, choose constant-1 local models along an exhaustion of G with errors tending to zero. Their classes define a homomorphism into $\prod_n \mathcal{U}(r_n)/\mathcal{N}$; eventual separation by 1 makes it injective. Lemma 2.2 identifies this quotient with $\mathcal{U}(\mathcal{Q}((r_n)))$. □

Theorem 2.5 (the MF locus is closed)**.** *For each $k \geq 1$, the MF groups form a closed subset of the space of k-marked groups. Equivalently, every non-MF k-marked group has a finite-radius obstruction: all marked groups agreeing with it on a sufficiently large word ball are non-MF.*

Proof. By Proposition 2.3, failure of MF is witnessed by a finite set and a positive tolerance for which the constant-1 local model does not exist. Choose words representing this finite set and all products appearing in its multiplicativity conditions. The relevant equalities and inequalities among these words determine a basic clopen neighborhood, and the same local obstruction holds throughout it. A word ball containing all of these words gives the finite-radius formulation. □

Directed-colimit permanence. The class of countable MF groups is closed under directed colimits.

Proof. Let $G = \varinjlim_i G_i$ with every G_i MF. Given finite $F \subseteq G$ and $\varepsilon > 0$, choose a stage containing lifts of all elements and products occurring in the local conditions on F, and then a later stage at which every equality and inequality among those finitely many elements has stabilized to its truth value in G. Composing a constant-1 local model at that stage with the lifts gives one with the prescribed data on F. Proposition 2.3 gives that G is MF. □

The amplification step plays the role that rank amplification plays in the direct-limit theorem for linear sofic groups [AP].

Finite-presentation reduction. The following are equivalent:

(1) a countable non-MF group exists;
(2) a finitely generated non-MF group exists;
(3) a finitely presented non-MF group exists.

Proof. If G is non-MF, a failed local condition is witnessed on a finite set $F \subseteq G$. The subgroup $H = \langle F \rangle$ is non-MF, since an MF model of H on that finite data would contradict the obstruction in G.

Write $H = F_k/N$. By Theorem 2.5, a basic cylinder around H, determined by finitely many words $W \subseteq F_k$, consists entirely of non-MF groups. Let R_+

be the words in W which are trivial in H, and put $P = F_k/\langle\langle R_+\rangle\rangle$. Then P is finitely presented and maps onto H. Every required equality holds in P, while every word of W nontrivial in H remains nontrivial in P because of the quotient map $P \twoheadrightarrow H$. Thus P lies in the same non-MF cylinder. The remaining implications are immediate. $\square$

Two stronger conventions also occur in the literature. The maps in [GKEMP] must recover the canonical trace and the left-regular norm of every group-ring element. The purely matricial field (PMF) property of [MdlS, Definition 1.2] requires finite-dimensional representations whose group-ring norms converge to the regular norms.

3. One-sided conjugation in matrix models

One-sided conjugation has analogous consequences in finite groups, finite-dimensional representations, and norm ultraproducts. For a homomorphism π to a finite group, $\pi(t\Gamma t^{-1}) = \pi(\Gamma)$, since the left side is a conjugate of the right side contained in it and the two are finite of the same cardinality. In finite dimensions, the commutant C of the base satisfies $C \subseteq \pi(t)C\pi(t)^{-1}$; equality follows because the two spaces have the same dimension. In a norm ultraproduct, the fixed-space projection satisfies $P \leq VPV^*$ when $L \leq H$ and $tLt^{-1} \subseteq L$, where P is the projection onto the L-fixed vectors and V represents t; the two projections are Murray–von Neumann equivalent, and finiteness of the ultraproduct gives $P = VPV^*$.

The two lemmas of this section are standard facts about finite C^*-algebras; see, e.g., [Bla].

Lemma 3.1 (comparison in a finite algebra)**.** *Let A be a finite unital C^*-algebra and let $p, q \in A$ be projections with $p \leq q$ and $p \sim q$. Then $p = q$.*

Proof. Let $r \in A$ satisfy $r^*r = q$ and $rr^* = p$. From $r^*r = q$ we get $r = rq$. Since $p \leq q$ we have $qp = p$, so

$$pr = rr^*r = rq = r, \qquad qr = q(pr) = (qp)r = pr = r,$$

and $r = qr$ as well. Hence the cross terms in $\sigma^*\sigma$ vanish for $\sigma = r + (1-q)$, and $\sigma^*\sigma = q + (1-q) = 1$, so σ is an isometry. Since A is finite, σ is unitary, and $\sigma\sigma^* = p + (1-q) = 1$ gives $p = q$. $\square$

Lemma 3.2 (finiteness of a norm quotient of matrix blocks)**.** *Let (K_n) be finite-dimensional Hilbert spaces and let*

$$B_{\mathrm{c}} = \prod_n B(K_n) \Big/ \bigoplus_n B(K_n)$$

be the quotient of the bounded family algebra by the c_0-sum. Then B_{c} is a finite C^*-algebra: every $\sigma \in B_{\mathrm{c}}$ with $\sigma^*\sigma = 1$ satisfies $\sigma\sigma^* = 1$.

Proof. Lift σ to a bounded family (σ_n). Then $\|\sigma_n^*\sigma_n - 1\| \to 0$, so for all large n the Gram defect is at most $\frac{1}{2}$ and $\sigma_n^*\sigma_n$ is invertible; each σ_n acts on a finite-dimensional space, so polar correction replaces it by a unitary

$w_n = \sigma_n(\sigma_n^*\sigma_n)^{-1/2}$ with $\|\sigma_n - w_n\| \to 0$. Hence σ is the class of (w_n), and $w_n w_n^* = 1$ at every such n gives $\sigma\sigma^* = 1$. □

Let ω be a free ultrafilter on $\mathbb{N}$, let K_ω be the Hilbert-space ultraproduct of the K_n, and let $B_\omega = \prod_\omega B(K_n)$ be the norm ultraproduct, acting on K_ω by $[A_n]_\omega[\xi_n]_\omega = [A_n\xi_n]_\omega$. The same polar-correction argument shows that B_ω is finite, with ordinary eventual convergence replaced by convergence along ω. Its action on K_ω is faithful: if $A = [A_n]_\omega \neq 0$ then $\lim_\omega \|A_n\| = \delta > 0$, and unit vectors ξ_n with $\|A_n\xi_n\| \geq \|A_n\| - 1/n$ give $\|A[\xi_n]_\omega\| = \delta$. Hence for projections $P, Q \in B_\omega$ the inclusion $\operatorname{ran} P \subseteq \operatorname{ran} Q$ makes QP and P act identically on K_ω, so $QP = P$, that is, $P \leq Q$; the converse is immediate.

The exact case.

Theorem 3.3 (the finite-dimensional case)**.** *Let k be a field, V a finite-dimensional k-vector space, H a group, $\Gamma \leq H$ a subgroup, $t, c \in H$, and $a \in \Gamma$. Assume that $t\Gamma t^{-1} \subseteq \Gamma$ and $c\gamma = \gamma c$ for all $\gamma \in \Gamma$. Then every homomorphism $\pi\colon H \to \mathrm{GL}(V)$ satisfies*

$$\pi\big([\,tct^{-1},\ a(tct^{-1})a^{-1}]\big) \;=\; 1.$$

Proof. Let

$$C = \{x \in \mathrm{End}_k(V) : \pi(\gamma)x = x\pi(\gamma) \text{ for all } \gamma \in \Gamma\}$$

be the commutant of $\pi(\Gamma)$, a linear subspace of $\mathrm{End}_k(V)$, and write $\Phi(x) = \pi(t)\,x\,\pi(t)^{-1}$ for conjugation by $\pi(t)$, a linear automorphism of $\mathrm{End}_k(V)$.

We claim $C \subseteq \Phi(C)$. Let $x \in C$ and put $y = \pi(t)^{-1}x\,\pi(t)$, so that $x = \Phi(y)$. For $\gamma \in \Gamma$ we have $t\gamma t^{-1} \in \Gamma$, hence $\pi(t\gamma t^{-1})$ commutes with x, and therefore

$$\pi(\gamma)\,y = \pi(t)^{-1}\,\pi(t\gamma t^{-1})\,x\,\pi(t) = \pi(t)^{-1}\,x\,\pi(t\gamma t^{-1})\,\pi(t) = y\,\pi(\gamma).$$

Thus $y \in C$, proving the claim. Since Φ is a linear isomorphism, $\dim_k \Phi(C) = \dim_k C < \infty$, and the inclusion $C \subseteq \Phi(C)$ forces

$$C \;=\; \Phi(C) \;=\; \pi(t)\,C\,\pi(t)^{-1}. \tag{3}$$

Now $\pi(c) \in C$ because c centralizes Γ, so $\pi(tct^{-1}) = \Phi(\pi(c)) \in C$ by (3). Since $a \in \Gamma$, the operator $\pi(tct^{-1})$ commutes with $\pi(a)$, so $\pi\big(a(tct^{-1})a^{-1}\big) = \pi(tct^{-1})$ and

$$\pi\big([\,tct^{-1},\ a(tct^{-1})a^{-1}]\big) = [\,\pi(tct^{-1}),\ \pi(tct^{-1})] = 1.$$ □

Every finite-dimensional linear representation of the group E of Definition 6.1 below, over any field, maps w to the identity; consequently every homomorphism from E to a finite group does as well.

In an MF model, multiplicativity is only asymptotic, so the coordinate fixed spaces need not be invariant. Because the multiplicative defects tend to 0 in operator norm, the adjoint actions on the Hilbert–Schmidt spaces are asymptotically multiplicative, and the corresponding Kazhdan projection belongs to a finite norm ultraproduct, where $P \leq VPV^*$ and $P \sim VPV^*$

imply $P = VPV^*$. The passage from exact representations to asymptotic ones is precisely where property (T) enters.

The asymptotic case. Recall that a *Kazhdan pair* for a property-(T) group L is a finite set $S \subseteq L$ together with a constant $\kappa > 0$ such that, in every unitary representation π of L with no nonzero invariant vector, every unit vector ξ satisfies $\|\pi(a)\xi - \xi\| \geq \kappa$ for some $a \in S$. The next lemma is the spectral-gap form of the Kazhdan projection, which goes back to Akemann and Walter [AW]; see [DN] for a general treatment.

Lemma 3.4 (Kazhdan spectral gap)**.** *Let L be a group with property* (T)*, let $S \subseteq L$ be a finite symmetric Kazhdan set containing* 1 *with Kazhdan constant $\kappa \in (0, 1]$, and let π be a unitary representation of L in a unital C^*-algebra. Then the average $h = \frac{1}{|S|} \sum_{a \in S} \pi(a)$ is self-adjoint and*

$$\mathrm{sp}(h) \subseteq \left[-1,\ 1 - \frac{\kappa^2}{2|S|}\right] \cup \{1\}.$$

Proof. Symmetry of S makes h self-adjoint, and $\|h\| \leq 1$ puts its spectrum in $[-1, 1]$. Let $\mu \in \mathrm{sp}(h)$ with $\mu \neq 1$. Evaluation at μ is a state on $C^*(1, h)$; extend it to a state φ on the ambient C^*-algebra. Let σ_φ be its GNS representation and set $\widetilde{\pi} = \sigma_\varphi \circ \pi$. Since $\varphi((h - \mu)^2) = 0$,

$$\|(\sigma_\varphi(h) - \mu)\xi_\varphi\|^2 = \varphi((h - \mu)^2) = 0.$$

Thus the cyclic vector $\xi = \xi_\varphi$ is a unit vector with $\sigma_\varphi(h)\xi = \mu\xi$. Split $\xi = \xi_0 + \xi_1$ along the $\widetilde{\pi}(L)$-invariant vectors and their orthocomplement, both invariant under $\sigma_\varphi(h)$. On the first summand $\sigma_\varphi(h) = 1$, so $\mu \neq 1$ forces $\xi_0 = 0$. Now

$$1 - \mu = \frac{1}{|S|} \sum_{a \in S} \big(1 - \mathrm{Re}\langle \widetilde{\pi}(a)\xi, \xi\rangle\big) = \frac{1}{2|S|} \sum_{a \in S} \|\widetilde{\pi}(a)\xi - \xi\|^2,$$

and $\widetilde{\pi}$ has no nonzero invariant vector on the orthocomplement, so $\|\widetilde{\pi}(a)\xi - \xi\| \geq \kappa$ for some $a \in S$ and $1 - \mu \geq \kappa^2/(2|S|)$. ☐

Theorem 3.5 (one-sided conjugation preserves asymptotic commutants)**.** *Let H be a group, let $L \leq H$ have property* (T)*, let $t \in H$, and let (d_n) be a sequence of positive integers. Suppose*

$$tLt^{-1} \subseteq L.$$

Let $U_n \colon H \to \mathcal{U}(d_n)$ be an operator-norm asymptotic representation. If (x_n) is uniformly operator-norm bounded and asymptotically centralizes L in normalized Hilbert–Schmidt norm,

$$\|[x_n, U_n(\ell)]_-\|_2 \longrightarrow 0 \qquad (\ell \in L),$$

then the conjugated sequence does as well:

$$\|[U_n(t)x_nU_n(t)^*, U_n(\ell)]_-\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Proof. Suppose the conclusion fails: there are $\ell_0 \in L$, $\delta > 0$, and an infinite set $I \subseteq \mathbb{N}$ with $\|[U_n(t)x_nU_n(t)^*, U_n(\ell_0)]_-\|_2 \geq \delta$ for $n \in I$. Fix a free ultrafilter ω on $\mathbb{N}$ with $I \in \omega$.

The adjoint model. Regard $M_{d_n}(\mathbb{C})$ as the Hilbert space $K_n = L^2(M_{d_n}(\mathbb{C}), \mathrm{tr}_{d_n})$ and let each group element act on it by conjugation, $\operatorname{Ad} U_n(g)\,\xi = U_n(g)\,\xi\,U_n(g)^*$. The operator-norm almost multiplicativity of U_n implies that $g \mapsto \operatorname{Ad} U_n(g)$ is an operator-norm asymptotic representation on these Hilbert spaces.

The ultraproduct. Let K_ω and $B_\omega = \prod_\omega B(K_n)$ be as in Section 3; write $[\,\cdot\,]_\omega$ for classes in either. The action of B_ω on K_ω is faithful, and B_ω is finite: every isometry in it is unitary, by the ultrafilter version of Lemma 3.2. For projections in B_ω, inclusion of ranges in K_ω is equivalent to the projection order. Asymptotic multiplicativity implies that

$$\pi\colon H \longrightarrow \mathcal{U}(B_\omega), \qquad \pi(g) = [\operatorname{Ad} U_n(g)]_\omega,$$

a homomorphism.

The projection onto the fixed subspace. Choose a finite symmetric Kazhdan set $S \subseteq L$ containing 1, with Kazhdan constant $\kappa \in (0, 1]$, put $h = \frac{1}{|S|}\sum_{s' \in S} \pi(s')$, and let $\mathrm{Fix} \subseteq K_\omega$ be the subspace of vectors fixed by every $\pi(\ell)$, $\ell \in L$. By Lemma 3.4, 1 is isolated in $\mathrm{sp}(h)$, so the spectral projection P of h at 1 belongs to B_ω by continuous functional calculus; since $h = 1$ exactly on Fix, its range is Fix.

The two projections agree. Let $V = \pi(t)$ and $Q = VPV^*$. Since $tLt^{-1} \subseteq L$, every vector of Fix is fixed by each $\pi(t\ell t^{-1})$, $\ell \in L$; and η is fixed by all $\pi(t\ell t^{-1})$, $\ell \in L$, exactly when $V^*\eta \in \mathrm{Fix}$. Hence $\operatorname{ran} P = \mathrm{Fix} \subseteq V\,\mathrm{Fix} = \operatorname{ran} Q$, so $P \leq Q$. The element $r = V^*Q$ satisfies $r^*r = Q$ and $rr^* = V^*QV = P$, so $P \sim Q$. Since B_ω is finite, Lemma 3.1 gives $Q = P$.

Conclusion. Let $\xi = [x_n]_\omega \in K_\omega$; the uniform operator-norm bound makes this class well defined. By unitary invariance of the normalized Hilbert–Schmidt norm, the commutator hypothesis says exactly that each $\pi(\ell)$, $\ell \in L$, fixes ξ. Thus $\xi \in \mathrm{Fix}$. Since $Q = P$, we obtain $V\xi \in V\,\mathrm{Fix} = \operatorname{ran} Q = \operatorname{ran} P = \mathrm{Fix}$. The matrix $U_n(t)x_nU_n(t)^*$ is the vector $\operatorname{Ad} U_n(t)\,x_n$ of K_n, so, again by unitary invariance,

$$\|[U_n(t)x_nU_n(t)^*, U_n(\ell)]_-\|_2 = \big\|(\operatorname{Ad} U_n(\ell) - 1)\ \operatorname{Ad} U_n(t)\,x_n\big\| \longrightarrow 0$$

along ω, for every $\ell \in L$. For $\ell = \ell_0$ this contradicts $I \in \omega$. $\square$

4. The central involution obstruction

We now isolate the final obstruction. Its corner reduction is modeled on the operator–Hilbert–Schmidt compression argument of Bachner–Dogon–Lubotzky [BDL, Proposition 1.5]. Their proof uses operator–Hilbert–Schmidt stability to replace the compressed asymptotic representation by nearby genuine finite-dimensional representations. Here Theorem 3.5 replaces that stability step: on the compressed corner, it forces the word whose square is the involution to be Hilbert–Schmidt close to the identity.

Let w be a central involution of H and let Θ be a homomorphism to $\mathcal{U}(\mathcal{Q})$ with $\Theta(w) \neq 1$. Then $\Theta(w)$ is a self-adjoint unitary commuting with $\Theta(H)$, and $q = \frac{1}{2}(1 - \Theta(w))$ is a nonzero projection on whose corner $q\mathcal{Q}q$ the compression of $\Theta(w)$ is $-q$ exactly. The argument below compresses a matrix model of H to the corners cut out by coordinate lifts of such a projection.

Lemma 4.1 (compression to an asymptotically central corner)**.** *Let H be a group, let $V_{g,n} \in \mathcal{U}(d_n)$ be an operator-norm asymptotic representation of H, and let $q_n \in M_{d_n}(\mathbb{C})$ be nonzero projections with*

$$\|[q_n, V_{g,n}]_-\| \longrightarrow 0 \qquad (g \in H).$$

Write $r_n = \operatorname{rank} q_n \geq 1$ and identify $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$. Then there are unitaries $W_{g,n} \in \mathcal{U}(r_n)$ with

$$\|W_{g,n} - q_n V_{g,n} q_n\| \longrightarrow 0 \qquad (g \in H),$$

and $(W_{g,n})$ is an operator-norm asymptotic representation of H on the corners.

Proof. Fix g, abbreviate $e = q_n$, $a = V_{g,n}$ and $\delta = \|[e,a]_-\|$, and note that the unit of the corner is e. From $eae - ae = [e,a]_- e$ and $ea^* - a^*e = -\big([e,a]_-\big)^*$ we get $\|eae - ae\| \leq \delta$ and $\|ea^* - a^*e\| \leq \delta$, so

$$\|(eae)^*(eae) - e\| = \|ea^*(ea - ae)e\| \leq \delta,$$

since $a^*a = 1$. Once $2\delta \leq \frac{1}{2}$ the corner element eae is invertible in $M_{r_n}(\mathbb{C})$. Its polar correction $W_{g,n} = eae\,\big((eae)^*(eae)\big)^{-1/2}$ is unitary with $\|W_{g,n} - eae\| \leq 4\delta$ by the estimate in the proof of Lemma 2.1. Set $W_{g,n} = 1$ at the finitely many remaining indices. For $g, h \in H$ write $b = V_{h,n}$ and $\varepsilon = \|ab - V_{gh,n}\|$. Then

$$\|(eae)(ebe) - eV_{gh,n}e\| \leq \|ea(eb - be)e\| + \|e(ab - V_{gh,n})e\| \leq \|[e,b]_-\| + \varepsilon,$$

which tends to 0. Adding the three polar corrections gives

$$\|W_{g,n}W_{h,n} - W_{gh,n}\| \longrightarrow 0. \qquad \square$$

For the rounding, compression, and polar-correction implementation, we follow Bachner–Dogon–Lubotzky [BDL, Lemmas 2.2–2.3, Proposition 2.4, and the proof of Proposition 1.5]. Their involution-rounding lemma, in turn, is due to De Chiffre–Glebsky–Lubotzky–Thom [DGLT, Proposition 1.4]. The normalization of a distinguished central involution by passage to its negative eigenspace appears earlier in Slofstra [Slo17, Lemma 3.9] and Slofstra–Vidick [SV, proof of Proposition 2.7]. The finite normal case, Their use here is specifically the normalization step; the obstruction below is supplied by one-sided Kazhdan transport and the square relation.

Theorem 4.2 (central involution obstruction)**.** *Let H be a countable group, let $L \leq H$ have property* (T)*, let $t, c \in H$, and suppose*

$$tLt^{-1} \subseteq L, \qquad [c, L] = 1.$$

Put $\hat{c} = tct^{-1}$ and $u = [\hat{c}, a]$ for some $a \in L$. If $w := u^2$ is a nontrivial central involution of H, then every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ maps w to the identity, and H is not MF.

Proof. Let $\Theta\colon H \to \mathcal{U}(\mathcal{Q})$ be a homomorphism and suppose $\Theta(w) \neq 1$.

The (-1)*-spectral corner.* Choose coordinate unitary lifts $(V_{g,n})$ of Θ (Lemma 2.2) and put $p_n = \frac{1}{2}(1 + V_{w,n})$, the averaging operator of the two-element subgroup $F = \{1, w\}$. Because the lifts are asymptotically multiplicative and $w^2 = 1$, we have $\left\|V_{w,n}^2 - 1\right\| \to 0$ and hence $\left\|V_{w,n}^* - V_{w,n}\right\| \to 0$. Set

$$a_n = \tfrac{1}{2}(V_{w,n} + V_{w,n}^*), \qquad e_n = \tfrac{1}{2}(1 + a_n).$$

Then $e_n = e_n^*$, $\|e_n - p_n\| \to 0$, and $\|e_n^2 - e_n\| \to 0$. The spectrum of e_n therefore concentrates near $\{0, 1\}$, and spectral rounding at $\frac{1}{2}$ gives projections at operator-norm distance $o(1)$ from p_n. Since w is central, these projections asymptotically commute with every lift of $\Theta(H)$. Let q_n be their complements. If q_n vanished for all large n the averages would converge to 1 and $\Theta(w)$ would be 1; so $q_n \neq 0$ along infinitely many coordinates, which we retain and relabel by $\mathbb{N}$. Lemma 4.1 now gives an operator-norm asymptotic representation $(W_{g,n})$ of H on the corners $q_n M_{d_n}(\mathbb{C}) q_n \cong M_{r_n}(\mathbb{C})$. On those corners $q_n V_{w,n} q_n = -q_n + o(1)$, because q_n is the complement of a projection within $o(1)$ of $\frac{1}{2}(1 + V_{w,n})$, so

$$\|W_{w,n} + 1\| \longrightarrow 0.$$

By the standing corner-trace convention, all estimates below use tr_{r_n} and are independent of the ratio r_n/d_n.

The commutant on the corner. Since c centralizes L, the sequence $(W_{c,n})$ centralizes $(W_{\ell,n})$, $\ell \in L$, asymptotically in operator norm, hence in normalized Hilbert–Schmidt norm. By Theorem 3.5 applied to $x_n = W_{c,n}$, the conjugated sequence $(W_{\hat{c},n})$ lies in the same asymptotic commutant. Therefore $\|W_{u,n} - 1\|_2 \to 0$, and squaring gives $\left\|W_{u,n}^2 - 1\right\|_2 \to 0$.

The contradiction. The relation $w = u^2$ gives $\left\|W_{u,n}^2 - W_{w,n}\right\| \to 0$, while $W_{w,n} \to -1$ in operator norm and therefore also in normalized Hilbert–Schmidt norm. Hence $\left\|W_{u,n}^2 + 1\right\|_2 \to 0$ and $\left\|W_{u,n}^2 - 1\right\|_2 \to 0$. The triangle inequality would then give $\|2 \cdot 1\|_2 \to 0$, contradicting $\|2 \cdot 1\|_2 = 2$. Thus $\Theta(w) = 1$. Since $w \neq 1$, every homomorphism $H \to \mathcal{U}(\mathcal{Q})$ has nontrivial kernel, so H is not MF. □

5. The basic non-MF construction

Theorem 5.1 (the basic construction)**.** *Let Γ be a finitely presented group with property* (T)*, let $\alpha\colon \Gamma \hookrightarrow \Gamma$ be an injective endomorphism, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Define $E(\Gamma, \alpha, a)$ by adjoining elements t, c and imposing*

$$t\gamma t^{-1} = \alpha(\gamma), \qquad c^2 = 1, \qquad [c, \gamma] = 1 \quad (\gamma \in \Gamma).$$

Put

$$\hat{c} = tct^{-1}, \qquad u = [\hat{c}, a], \qquad w = u^2 = [\hat{c}, a\hat{c}a^{-1}],$$

and impose centrality of w. More explicitly, for a finite generating set S_Γ it is enough to impose the conjugation and commutation relations for $s \in S_\Gamma$, together with

$$[w, s] = 1 \quad (s \in S_\Gamma), \qquad [w, t] = [w, c] = 1.$$

The group $E(\Gamma, \alpha, a)$ is finitely presented, the canonical map $\Gamma \to E(\Gamma, \alpha, a)$ is injective, and w is a nontrivial central involution. Every homomorphism $E(\Gamma, \alpha, a) \to \mathcal{U}(\mathcal{Q})$ maps w to 1, so $E(\Gamma, \alpha, a)$ is not MF.

Proof. Finite presentability follows from the relative presentation. Start with the free product of Γ and the free group on t, c, and quotient by the finitely many displayed relations for a finite generating set of Γ.

The relation $w^2 = 1$ need not be imposed. Put $b = a\hat{c}a^{-1}$. Both $\hat{c}$ and b are involutions, $u = [\hat{c}, a] = \hat{c}b$, and $w = u^2 = (\hat{c}b)^2$. Since $\hat{c}^2 = 1$,

$$\hat{c}\, w\, \hat{c}^{-1} = (b\hat{c})^2 = w^{-1}. \tag{4}$$

Centrality also gives $\hat{c}w\hat{c}^{-1} = w$, whence $w = w^{-1}$ and $w^2 = 1$.

For nontriviality, consider the ascending HNN extension

$$G = \langle \Gamma, t \mid t\gamma t^{-1} = \alpha(\gamma) \rangle$$

and use Britton's lemma to identify Γ with its canonical copy in G.

Let $X = G/\Gamma$ be its coset space. The Clifford group $\mathrm{Cl}(X)$ is generated by a central involution ζ and involutions c_x, $x \in X$, subject to $[c_x, c_y] = \zeta$ for $x \neq y$ (Construction 8.1). Every permutation of X acts on $\mathrm{Cl}(X)$ by permuting the c_x and fixing ζ. Let G act in this way and map c to c_Γ. Then

$$\hat{c} \longmapsto c_{t\Gamma}, \qquad a\hat{c}a^{-1} \longmapsto c_{at\Gamma}.$$

The two cosets are distinct: equality would give $t^{-1}at \in \Gamma$, equivalently $a \in \alpha(\Gamma)$. Hence the two Clifford generators anticommute and

$$w \longmapsto (c_{t\Gamma} c_{at\Gamma})^2 = \zeta \neq 1.$$

The sign ζ is central, so the map respects every defining relation. It follows both that $w \neq 1$ and that the canonical map of Γ is injective.

Thus $w = u^2$ is a nontrivial central involution, and by Theorem 4.2 every homomorphism to $\mathcal{U}(\mathcal{Q})$ maps w to the identity. Since $w \neq 1$, every such homomorphism has nontrivial kernel. □

Finite presentability of Γ is used only to make $E(\Gamma, \alpha, a)$ finitely presented; for the rest, Γ may be any property-(T) group with a proper injective endomorphism.

6. The explicit affine example

Specialize Theorem 5.1 to

$$\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}), \qquad \alpha(v, A) = (2v, A), \qquad a = (e_1, I). \tag{5}$$

The affine group Γ is finitely presented, and it has property (T) because it is a lattice in $\mathbb{R}^3 \rtimes \mathrm{SL}_3(\mathbb{R})$ [BHV, Corollary 1.4.16 and Theorem 1.7.1]; see also [Bu]. Moreover, Section 7 identifies it with the presentation used below. The endomorphism α is injective because doubling is injective on $\mathbb{Z}^3$, and it is not surjective because $e_1 \notin 2\mathbb{Z}^3$. The same observation gives $a \notin \alpha(\Gamma)$. Hence Theorem 5.1 applies. Relative to Γ, the resulting group has the finite presentation

$$E = \left\langle \Gamma, t, c \;\middle|\; \begin{array}{c} t\gamma t^{-1} = \alpha(\gamma), \quad [c, \gamma] = 1 \quad (\gamma \in S_\Gamma), \\ c^2 = 1, \qquad [w, g] = 1 \end{array} \right\rangle,$$

where S_Γ is a finite generating set of Γ, g runs over the generators, and $w = [\,tct^{-1},\, a\,(tct^{-1})\,a^{-1}]$. Substituting a standard presentation of the affine group in the generators v_1, v_2, v_3 for the translations and x, y, z for the linear part yields the eight-generator, forty-one-relator presentation below. First define the *linear presentation*

$$\mathcal{R} = \left\langle x, y, z \;\middle|\; \begin{array}{c} x^3 = y^3 = z^2 = (xz)^3 = (yz)^3 = 1, \\ (x^{-1}zxy)^2 = (y^{-1}zyx)^2 = (xy)^6 = 1 \end{array} \right\rangle, \tag{6}$$

and the *base*

$$\mathcal{B} = \left\langle v_1, v_2, v_3, x, y, z \;\middle|\; \begin{array}{c} \text{the eight relations in (6)}, \\ [v_1, v_2] = [v_1, v_3] = [v_2, v_3] = 1, \\ xv_1x^{-1} = v_3,\ xv_2x^{-1} = v_1,\ xv_3x^{-1} = v_2, \\ yv_1y^{-1} = v_1,\ yv_2y^{-1} = v_2^{-1}v_3,\ yv_3y^{-1} = v_1v_2^{-1}, \\ zv_1z^{-1} = v_2v_3^{-1},\ zv_2z^{-1} = v_1v_3^{-1},\ zv_3z^{-1} = v_3^{-1} \end{array} \right\rangle. \tag{7}$$

Write $\iota\colon \mathcal{B} \to E$ for the homomorphism induced by the first six generators of the presentation below.

Definition 6.1. Use the six base generators and the relations displayed in (7). Define

$$E = \left\langle v_1, v_2, v_3, x, y, z, t, c \;\middle|\; \begin{array}{c} \text{(B)} \quad \text{all relations in (7)}, \\ \text{(T)} \quad tv_it^{-1} = v_i^2 \ (1 \le i \le 3), \\ txt^{-1} = x, \quad tyt^{-1} = y, \quad tzt^{-1} = z, \\ \text{(C)} \quad c^2 = 1, \\ [c, g] = 1 \ \ (g \in \{v_1, v_2, v_3, x, y, z\}), \\ \text{(W)} \quad [w, g] = 1 \quad (g \in \{v_1, v_2, v_3, x, y, z, t, c\}) \end{array} \right\rangle, \tag{8}$$

where w abbreviates the word $[tct^{-1}, v_1(tct^{-1})v_1^{-1}]$. This is an explicit finite presentation with eight generators and forty-one relators: twenty in family (B), six in family (T), seven in family (C), and eight in family (W). Centrality of w together with $c^2 = 1$ implies $w^2 = 1$ by (4). Families (T) and (C) give, respectively,

$$t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B}), \qquad [c, \iota(\mathcal{B})] = 1.$$

Figure 1 shows how the relation families enter the obstruction and the nontriviality witness.

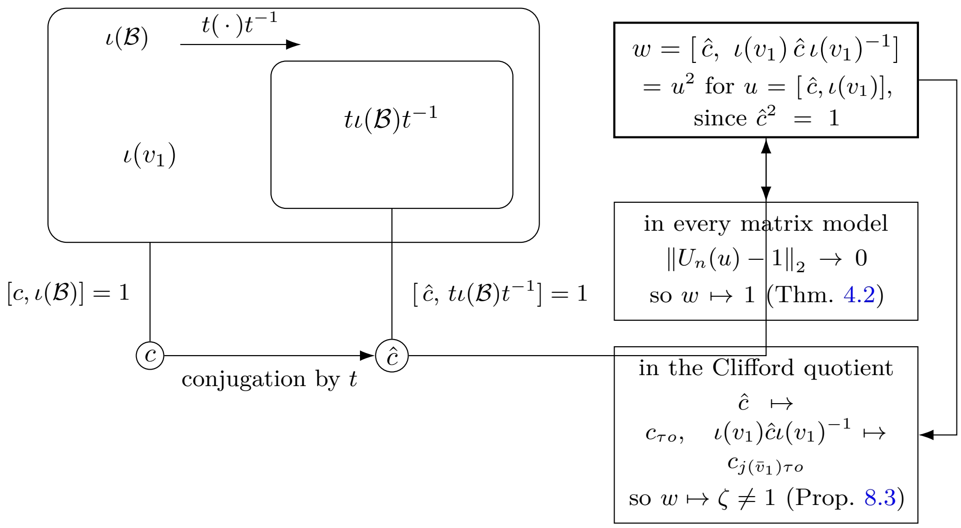

Figure 1: The relations of (8) give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, hence $[\hat{c}, t\iota(\mathcal{B})t^{-1}] = 1$ for $\hat{c} = tct^{-1}$. Every homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$ has $\Theta(w) = 1$; the Clifford quotient of Section 8 maps w to $\zeta \neq 1$.

Lemma 6.2 (the distinguished word is a square)**.** *Let H_1 be a group and $x, y \in H_1$ with $x^2 = 1$. Then*

$$[x,\ y\,x\,y^{-1}] \;=\; [x, y]^2.$$

Proof. Put $z = yxy^{-1}$, so $z^2 = yx^2y^{-1} = 1$; thus $z^{-1} = z$ and $x^{-1} = x$. Then $[x, y] = xyx^{-1}y^{-1} = x\,(yxy^{-1}) = xz$, and

$$[x, y]^2 = (xz)^2 = xz\,xz = x\,z\,x^{-1}z^{-1} = [x, z]. \qquad \square$$

Proof of Theorem A. The group E is the specialization of Theorem 5.1 to the data (5), and $w \neq 1$ by Proposition 8.3.

Let (d_n) be given, write $\mathcal{Q} = \mathcal{Q}((d_n))$, and let $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$ be a homomorphism. Apply the central involution obstruction (Theorem 4.2) with $L = \iota(\mathcal{B}) \leq E$, the elements $t, c \in E$, and $a = \iota(v_1)$. The basic construction identifies $\mathcal{B}$ with L, so Proposition 7.1 gives property (T) of L, while families (T) and (C) give $t\iota(\mathcal{B})t^{-1} \subseteq \iota(\mathcal{B})$ and $[c, \iota(\mathcal{B})] = 1$, respectively. In E we have $\hat{c}^2 = (tct^{-1})^2 = tc^2t^{-1} = 1$, so Lemma 6.2 with $x = \hat{c}$ and $y = \iota(v_1)$ gives

$$w = [\,\hat{c},\ \iota(v_1)\,\hat{c}\,\iota(v_1)^{-1}] = [\,\hat{c}, \iota(v_1)\,]^2 = u^2, \qquad u = [\,\hat{c}, \iota(v_1)\,].$$

The defining relations make w central, and $c^2 = 1$ then implies $w^2 = 1$ by (4). Hence $\Theta(w) = 1$.

Every homomorphism $E \to \mathcal{U}(\mathcal{Q})$ maps the nontrivial element w to 1, so none is injective and E is not MF. The canonical maps $E \to \mathcal{U}(C^*_{\mathrm{r}}(E))$ and $E \to \mathcal{U}(C^*_{\max}(E))$ are injective, the first because $g \mapsto \lambda_g$ is and the second because the left regular representation factors through $C^*_{\max}(E)$. If $e\colon A \to \mathcal{Q}$ is an injective, possibly nonunital, $*$-homomorphism from a unital algebra, then $u \mapsto e(u) + 1 - e(1)$ is an injective homomorphism $\mathcal{U}(A) \to \mathcal{U}(\mathcal{Q})$. Hence an MF embedding of either $C^*_{\mathrm{r}}(E)$ or $C^*_{\max}(E)$ would give an injective homomorphism $E \to \mathcal{U}(\mathcal{Q})$, a contradiction. $\square$

Corollary 6.3 (uniform finite operator-norm obstruction)**.** *There exist $\delta > 0$ and a finite set $F_0 \subset E$ such that, for every $d \geq 1$, every map $\phi\colon E \to \mathcal{U}(d)$ satisfying*

$$\|\phi(gh) - \phi(g)\phi(h)\| \leq \delta \qquad (g, h \in F_0)$$

satisfies $\|\phi(w) - 1\| < 1$.

Proof. Fix an exhaustion $F_1 \subseteq F_2 \subseteq \cdots$ of E by finite sets. If no such pair (δ, F_0) exists, then for each n there is a unitary-valued ϕ_n with multiplicative defect at most $1/n$ on F_n and $\|\phi_n(w) - 1\| \geq 1$. The sequence (ϕ_n) is an operator-norm asymptotic representation of E, so by Lemma 2.2 its class is a homomorphism $\Theta\colon E \to \mathcal{U}(\mathcal{Q})$, and $\|\Theta(w) - 1\| \geq 1$ gives $\Theta(w) \neq 1$, contradicting Theorem A. $\square$

Since E is finitely presented, the forty-one relators of (8) serve as the test set, and the hypothesis becomes a condition on an eight-tuple of unitaries.

Corollary 6.4 (a uniform obstruction on the presentation)**.** *There is a $\delta > 0$ with the following property. For every $d \geq 1$ and every eight-tuple $V_1, \ldots, V_8 \in \mathcal{U}(d)$, if*

$$\|r(V_1, \ldots, V_8) - 1\| \leq \delta \qquad \textit{for each of the forty-one relators } r \textit{ of } (8),$$

then $\|w(V_1, \ldots, V_8) - 1\| < 1$.

Proof. If no such δ exists, then for every n there are a dimension d_n and a tuple $V^{(n)}$ satisfying every relator to within $1/n$ and with $\left\|w(V^{(n)}) - 1\right\| \geq 1$. Each tuple defines a homomorphism $\varphi_n\colon F_8 \to \mathcal{U}(d_n)$, since F_8 is free, and by Lemma 2.2 the classes define a homomorphism $\Phi\colon F_8 \to \mathcal{U}(\mathcal{Q}((d_n)))$. Each of the forty-one relators lies in $\ker\Phi$, its images converging to 1 in operator norm, so Φ factors as $\Theta \circ \pi$ through the quotient map $\pi\colon F_8 \to E$.

Theorem A gives $\Theta(w) = 1$, that is $\left\| w(V^{(n)}) - 1 \right\| \to 0$, contradicting the lower bound 1. $\square$

The compactness proof is ineffective. An explicit value of δ and an explicit test set F_0 would require an effective Kazhdan constant for the base and effective compression estimates.

7. The base and a smaller presentation

The base $\mathcal{B}$ of (7) is the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, which has property (T) as a lattice in $\mathbb{R}^3 \rtimes \mathrm{SL}_3(\mathbb{R})$ [BHV, Corollary 1.4.16 and Theorem 1.7.1]; see also [Bu].

A six-generator presentation. The group E is generated by the six elements v_1, x, y, z, t, c, and has a Tietze-equivalent presentation on them with thirty-two relators. In (8), eliminate $v_3 = xv_1x^{-1}$ and $v_2 = x^2v_1x^{-2}$ by Tietze substitutions and delete the two defining relations. The remaining x-action relation follows from $x^3 = 1$. The stable-letter, c-commutation, and centrality relations for v_2, v_3 follow from those for v_1 together with $[t, x] = 1$ and $x^3 = 1$. Deleting these nine leaves thirty-two relators on v_1, x, y, z, t, c.

After these Tietze transformations, the remaining relations still imply $[\,\hat{c},\, t\iota(\mathcal{B})t^{-1}] = 1$ for $\hat{c} = tct^{-1}$, and the distinguished word $[\,\hat{c},\ \iota(v_1)\,\hat{c}\,\iota(v_1)^{-1}]$, computed in the same group, is unchanged.

7.1. Property (T) of the base.

Proposition 7.1. *The twenty-relator group $\mathcal{B}$ has property* (T)*, in both the real-orthogonal and complex-unitary formulations.*

Proof. By Remark 7.2 the presentation (7) defines $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, which has property (T) [BHV, Corollary 1.4.16 and Theorem 1.7.1]. Let ρ be an orthogonal representation on a real Hilbert space $H_{\mathbb{R}}$. Its complexification $H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ has the unitary representation $\rho \otimes 1$. The invariant vectors and displacement norms are those of ρ in each real coordinate, so every Kazhdan pair for the unitary formulation is also one for the orthogonal formulation. $\square$

Remark 7.2 (the classical identification). The eight relations (6) present $\mathrm{SL}_3(\mathbb{Z})$ [CRW, Theorem 2]; the corresponding matrices are also recorded in [CLV]. Hence $\mathcal{B} \cong \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$, and the surjection $\mathcal{B} \to \bar{\Gamma}$ of Lemma 8.2 onto the matrix realization is an isomorphism.

8. An auxiliary Clifford witness

Construction 8.1 (Clifford groups)**.** For a set X, use the real Clifford-algebra convention generated by $(c_x)_{x \in X}$ with $c_x^2 = 1$ and $c_xc_y = -c_yc_x$ for $x \neq y$, and let $\mathrm{Cl}(X)$ be the subgroup of units generated by -1 and the c_x. Write $\zeta = -1$. In this concrete group,

$$\zeta^2 = c_x^2 = 1, \qquad c_xc_y = \zeta c_yc_x \quad (x \neq y).$$

The sign ζ is nontrivial. Thus $\mathrm{Cl}(X)$ is a central extension of the elementary abelian 2-group on X, with central kernel generated by ζ. For distinct $x, y \in X$, the commutator $[c_x, c_y]$ equals ζ. Every permutation of X acts on this group by $c_x \mapsto c_{\sigma(x)}$ and fixes ζ.

Lemma 8.2 (linear model)**.** *For $M \in \mathrm{GL}_3(\mathbb{Q})$ and $u \in \mathbb{Q}^3$, write*

$$A(M,u) = \begin{pmatrix} M & u \\ 0 & 1 \end{pmatrix}.$$

Assign $v_i \mapsto A(I_3, e_i)$ and

$$x \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, 0\right), \qquad y \mapsto A\left(\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 0 & 1 & 0 \end{pmatrix}, 0\right),$$

$$z \mapsto A\left(\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{pmatrix}, 0\right).$$

Let $\bar{\Gamma} \le \mathrm{GL}_4(\mathbb{Q})$ be the subgroup generated by these six matrices, and put $D = \mathrm{diag}(2,2,2,1)$. The six matrices satisfy all twenty relators of $\mathcal{B}$, and

$$\bar{\alpha}(u) = DuD^{-1}$$

is an injective endomorphism of $\bar{\Gamma}$ which squares the three translation generators and fixes x, y, z. Moreover the matrix assigned to v_1 does not lie in $\mathrm{range}(\bar{\alpha})$.

Proof. Each of the twenty relators reduces to an equality between explicit 4×4 rational matrices. Conjugation by D gives the six stable-letter equations and is injective because it is conjugation by an invertible matrix. Every displayed generator and its explicitly computed inverse has integral entries; equivalently, all six generators lie in $\mathrm{GL}_4(\mathbb{Z})$. Hence $\bar{\Gamma} \le \mathrm{GL}_4(\mathbb{Z})$. However $D^{-1}\bar{v}_1 D$ has entry $1/2$ in position $(1,4)$, so it is not in $\bar{\Gamma}$. This proves the last assertion. ☐

Form the direct limit $T = \varinjlim(\bar{\Gamma}, \bar{\alpha})$ and let τ denote its shift automorphism. Set $\Sigma = T \rtimes \mathbb{Z}$. This is the group G used in Theorem 5.1 for the data (5). Let $j\colon \bar{\Gamma} \hookrightarrow \Sigma$ be the level-zero map. Put $X = \Sigma / j(\bar{\Gamma})$, with base point $o = j(\bar{\Gamma})$. Since $D^{-1}\bar{v}_1 D \notin \bar{\Gamma}$ (Lemma 8.2), we have $\tau o \ne j(\bar{v}_1)\tau o$, and the Clifford generators at these two points anticommute.

Proposition 8.3. *$w \ne 1$ in E.*

Proof. Form $W := \mathrm{Cl}(X) \rtimes \Sigma$ as in Construction 8.1. Define a map on the generators of (8) by

$$g \mapsto j(\bar{g}) \in \Sigma \quad (g \in \{v_1, v_2, v_3, x, y, z\}),$$
$$t \mapsto \tau, \qquad c \mapsto c_o.$$

Lemma 8.2 gives the twenty base and six stable-letter relations. The level-zero subgroup fixes o, so c_o commutes with every base generator, and $c_o^2 = 1$. Under this assignment,

$$tct^{-1} \mapsto c_{\tau o}, \qquad v_1(tct^{-1})v_1^{-1} \mapsto c_{j(\bar{v}_1)\tau o},$$

and the two points are distinct by Lemma 8.2, so

$$w \mapsto [c_{\tau o}, c_{j(\bar{v}_1)\tau o}] = -1 = \zeta.$$

Since ζ is central in W, the remaining relations $[w, g] = 1$ for $g \in \{v_1, v_2, v_3, x, y, z, t, c\}$ are also satisfied. Hence the assignment extends to a group homomorphism $E \to W$ sending w to $\zeta \neq 1$, and therefore $w \neq 1$ in E. □

Further results

Further variants and consequences are collected in the companion *Non-MF group notes*. The notes preserve the extended developments without interrupting the central-sign proof given here.

Acknowledgements

We thank Caleb Eckhardt for discussions clarifying the finite-quotient analogy that organizes Section 3. We thank Francesco Fournier-Facio for emphasizing the formulation in terms of a non-co-Hopfian Kazhdan group and its ascending HNN extension, for suggesting the small-cancellation construction recorded in the companion notes, and for helpful discussions about the operator and Hilbert–Schmidt norms. We thank Alon Dogon for helpful discussions and for organizing the early expert feedback on the manuscript.

AI and computational resource usage

Large language models were used extensively over multiple days and thousands of prompts for mathematical exploration, proof development, Lean formalization, programming, and editing. A custom informal proof tool supplied additional scaffolding. GPT-5.6 Sol first formulated the non-MF construction after the author suggested applying the existing work to the MF problem instead of the hyperlinear problem. Claude Fable 5 and Claude Opus 5 also contributed to proof development and formalization. The author checked and revised the resulting arguments and citations. A supercomputer and other computational experiments were used to test and discard preliminary approaches; the proofs in this paper do not depend on those computations.

References

[AW] C. A. Akemann and M. E. Walter, *Unbounded negative definite functions*, Canad. J. Math. **33** (1981), no. 4, 862–871.

[AT] V. Alekseev and A. Thom, *Centralizers of sofic approximations of Kazhdan groups*, preprint, 2026, arXiv:2608.05362.

[AP] G. Arzhantseva and L. Păunescu, *Linear sofic groups and algebras*, Trans. Amer. Math. Soc. **369** (2017), no. 4, 2285–2310, doi:10.1090/tran/6719.

[BDL] B. Bachner, A. Dogon, and A. Lubotzky, *On L^1-approximation of groups*, J. Algebra **702** (2026), 235–243, doi:10.1016/j.jalgebra.2026.04.041.

[Be] M. E. B. Bekka, *Amenable unitary representations of locally compact groups*, Invent. Math. **100** (1990), no. 2, 383–401, doi:10.1007/BF01231192.

[BV] M. E. B. Bekka and A. Valette, *Kazhdan's property (T) and amenable representations*, Math. Z. **212** (1993), no. 2, 293–299, doi:10.1007/BF02571659.

[BHV] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan's property* (T), New Mathematical Monographs, vol. 11, Cambridge University Press, Cambridge, 2008, doi:10.1017/CBO9780511542749.

[Bla] B. Blackadar, *Operator algebras: theory of C^*-algebras and von Neumann algebras*, Encyclopaedia of Mathematical Sciences, vol. 122, Springer, Berlin, 2006.

[BK] B. Blackadar and E. Kirchberg, *Generalized inductive limits of finite-dimensional C^*-algebras*, Math. Ann. **307** (1997), no. 3, 343–380, doi:10.1007/s002080050039.

[Bu] M. Burger, *Kazhdan constants for* $\mathrm{SL}_3(\mathbb{Z})$, J. Reine Angew. Math. **413** (1991), 36–67.

[CDE] J. R. Carrión, M. Dadarlat, and C. Eckhardt, *On groups with quasidiagonal C^*-algebras*, J. Funct. Anal. **265** (2013), no. 1, 135–152, doi:10.1016/j.jfa.2013.04.004, arXiv:1210.4050.

[CLV] M. D. E. Conder, G. Liversidge, and M. R. Vsemirnov, *Generating pairs for* $\mathrm{SL}(n,\mathbb{Z})$, J. Algebra **662** (2025), 123–137, doi:10.1016/j.jalgebra.2024.08.008.

[CRW] M. D. E. Conder, E. F. Robertson, and P. Williams, *Presentations for 3-dimensional special linear groups over integer rings*, Proc. Amer. Math. Soc. **115** (1992), no. 1, 19–26, doi:10.2307/2159559.

[Dad] M. Dadarlat, *Obstructions to matricial stability of discrete groups and almost flat K-theory*, Adv. Math. **384** (2021), Paper No. 107722, doi:10.1016/j.aim.2021.107722.

[DGLT] M. De Chiffre, L. Glebsky, A. Lubotzky, and A. Thom, *Stability, cohomology vanishing, and nonapproximable groups*, Forum Math. Sigma **8** (2020), Paper No. e18, doi:10.1017/fms.2020.5.

[DN] C. Druţu and P. W. Nowak, *Kazhdan projections, random walks and ergodic theorems*, J. Reine Angew. Math. **754** (2019), 49–86, doi:10.1515/crelle-2017-0002.

[FFF] F. Fournier-Facio, *A torsion-free non-sofic group*, preprint, arXiv:2608.02025; cited in version v2 (14 August 2026).

[FGH] A. Fox, I. Goldbring, and B. Hart, *Locally universal C^*-algebras with computable presentations*, J. Funct. Anal. **287** (2024), no. 12, Paper No. 110652, doi:10.1016/j.jfa.2024.110652, arXiv:2303.02301.

[GKEMP] D. Gao, S. Kunnawalkam Elayavalli, A. Manzoor, and G. Patchell, *A new source of purely finite matricial fields*, preprint, 2026, arXiv:2603.24502.

[HT] U. Haagerup and S. Thorbjørnsen, *A new application of random matrices:* $\mathrm{Ext}(C^*_{\mathrm{red}}(F_2))$ *is not a group*, Ann. of Math. (2) **162** (2005), no. 2, 711–775, doi:10.4007/annals.2005.162.711, arXiv:math/0212265.

[JNVWY] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen, MIP* = RE, Commun. ACM **64** (2021), no. 11, 131–138, doi:10.1145/3485628, arXiv:2001.04383.

[Ku16] G. Kun, *On sofic approximations of property (T) groups*, preprint, 2016, arXiv:1606.04471.

[KT19] G. Kun and A. Thom, *Inapproximability of actions and Kazhdan's property (T)*, preprint, 2019, arXiv:1901.03963.

[KT26] G. Kun and A. Thom, *Nonsofic wreath products of residually finite groups*, preprint, 2026, arXiv:2608.06222.

[Lo98] T. A. Loring, *C^*-algebras that are only weakly semiprojective*, Proc. Amer. Math. Soc. **126** (1998), no. 9, 2713–2715, doi:10.1090/S0002-9939-98-04292-0.

[LuO] A. Lubotzky and I. Oppenheim, *Non p-norm approximated groups*, J. Anal. Math. **141** (2020), no. 1, 305–321, doi:10.1007/s11854-020-0119-2, arXiv:1807.06790.

[MdlS] M. Magee and M. de la Salle, $\mathrm{SL}_4(\mathbb{Z})$ *is not purely matricial field*, C. R. Math. Acad. Sci. Paris **362** (2024), 903–910, doi:10.5802/crmath.617, arXiv:2312.03220.

[OAI] OpenAI, *Nonsofic groups exist*, in *Ten advances in mathematics and theoretical computer science*, Chapter 3, 78–95, 2026, https://cdn.openai.com/pdf/ten-proofs-oai.pdf.

[Sc] C. Schafhauser, *Finite-dimensional approximations of certain amalgamated free products of groups*, Groups Geom. Dyn. **20** (2026), no. 2, 607–615, doi:10.4171/GGD/826, arXiv:2306.02498.

[Sc20] C. Schafhauser, *Subalgebras of simple AF-algebras*, Ann. of Math. (2) **192** (2020), no. 2, 309–352, doi:10.4007/annals.2020.192.2.1, arXiv:1807.07381.

[Sh] T. Shulman, *The MF property for amalgamated free products*, preprint, 2026, arXiv:2603.13564; cited in version v2 (22 March 2026).

[Slo17] W. Slofstra, *The set of quantum correlations is not closed*, Forum Math. Pi **7** (2019), Paper No. e1, doi:10.1017/fmp.2018.3.

[Slo18] W. Slofstra, *A group with at least subexponential hyperlinear profile*, preprint, 2018, arXiv:1806.05267.

[SV] W. Slofstra and T. Vidick, *Entanglement in non-local games and the hyperlinear profile of groups*, Ann. Henri Poincaré **19** (2018), no. 10, 2979–3005, doi:10.1007/s00023-018-0718-y.

[Th] A. Thom, *Finitary approximations of groups and their applications*, Proceedings of the International Congress of Mathematicians — Rio de Janeiro 2018, arXiv:1712.01052.

[TWW] A. Tikuisis, S. White, and W. Winter, *Quasidiagonality of nuclear C^*-algebras*, Ann. of Math. (2) **185** (2017), no. 1, 229–284, doi:10.4007/annals.2017.185.1.4, arXiv:1509.08318.